

	1	2	3	4	5	6	7	8		
A	GENERATED STARTER SCAFFOLD ONLY – HUMAN ELECTRICAL REVIEW REQUIRED; NOT ELECTRICALLY VERIFIED OR APPROVED FOR FABRICATION									
	Generated notes only: no symbols, nets, footprints, or pin mappings are approved.									
	BLE antenna: TODO: VERIFY exact ESP32–S3 module edge placement and datasheet keepout.									
	NFC antenna: TODO: VERIFY loop geometry, metal exclusions, tuning network, and assembled measurement.									
	PLACEMENT: keep 49 x 99 mm product, 46 x 84 mm PCB, upper display, and lower rear LiPo.									
B	PLACEMENT: review FPC direction, USB–C opening, buttons, RGB guides, piezo, motor, and pogo access.									
	TEST ACCESS: bare pogo pads only; no pin headers in the finished thin product.									
	SECTION: Power input and LiPo charger			SECTION: 240x320 TFT display connector			SECTION: Buttons			
	Candidate direction: USB–C 5 V input, protected LiPo, BQ24074–class power path			Candidate direction: 14–pin 0.5 mm FPC connector for ST7789–class SPI panel			Candidate direction: Three SKRSPACE010–class tactile switches			
	Review scope: Input limits, charge current, NTC strategy, status, and thermal path			Review scope: FPC contact side, SPI, reset/DC/CS, backlight, and insertion direction			Review scope: GPIO straps, pull–ups, debounce, wake behavior, and actuator direction			
C	TODO: VERIFY exact footprint and land pattern before symbol placement.			TODO: VERIFY exact footprint and land pattern before symbol placement.			TODO: VERIFY exact footprint and land pattern before symbol placement.			
	TODO: VERIFY datasheet–confirmed pinout and electrical limits.			TODO: VERIFY datasheet–confirmed pinout and electrical limits.			TODO: VERIFY datasheet–confirmed pinout and electrical limits.			
	TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual–install state.			TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual–install state.			TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual–install state.			
	SECTION: 3.3 V power rail			SECTION: Display backlight control			SECTION: Piezo buzzer			
	Candidate direction: TP563031–class buck–boost with inductor and decoupling			Candidate direction: Default–off PWM control with panel–specific current regulation			Candidate direction: PKLCS1212E4001–R1–class passive piezo			
D	Review scope: Peak load, stability, efficiency, brownout, and power sequencing			Review scope: LED string voltage/current, dimming, EMI, thermal behavior, and sleep state			Review scope: Drive topology, voltage, resonant frequency, acoustic port, and quiet state			
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	SECTION: LiPo battery connector			SECTION: Dynamic NFC tag and I2C			SECTION: Optional vibration motor driver			
E	Candidate direction: Protected single–cell LiPo and low–profile polarized connection			Candidate direction: ST25DV04KC–class dynamic tag, I2C pull–ups, and tuning placeholders			Candidate direction: DNP motor pads and AO3400A–class low–side switch			
	Review scope: Polarity, harness, strain relief, cable exit, protection, and swelling volume			Review scope: I2C address, GPO/FD, energy harvesting decision, loop, and measurement pads			Review scope: Stall current, 3.3 V gate drive, flyback, EMI, timeout, and DNP safety			
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F	SECTION: ESP32–S3 module			SECTION: NFC antenna loop and tuning			SECTION: Battery voltage measurement			
	Candidate direction: ESP32–S3–MINI–1–N4R2 direction from V1 BOM shortlist			Candidate direction: Perimeter loop, links, DNP tuning network, and measurement pads			Candidate direction: Low–leakage divider or switched measurement into an ESP32–S3 ADC			
	Review scope: Power, EN/BOOT, straps, GPIO allocation, USB, and decoupling			Review scope: BLE detour, assembled inductance/Q, matching, fixture access, and interoperability			Review scope: ADC range, source impedance, filtering, calibration, leakage, and protection			
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	SECTION: EN BOOT and strapping			SECTION: Optional SPI NOR flash			SECTION: Test pads			
	Candidate direction: Reviewed reset, boot–mode, recovery, and strap–safe GPIO network			Candidate direction: W25Q128JV–class 3.3 V SPI NOR; DNP until architecture review			Candidate direction: Bare ENIG pogo pads; no unverified SMT test–point footprints			
	Review scope: Power–up defaults, manual recovery, native USB, and test–pad access			Review scope: Bus sharing, chip select, hold/write protect, decoupling, and DNP safety			Review scope: GND, VBUS, VBAT, 3V3, EN, BOOT, UART, USB, I2C, SPI, and switched loads			
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	SECTION: USB–C / USB data / ESD			SECTION: RGB LEDs			No electrical verification or fabrication approval Starter/scaffold requiring human electrical review MCard–StarterKit clean–room hardware study Sheet: / File: MCard_V1.kicad_sch Title: MCard V1 generated schematic scaffold Size: A3Date:Rev: KiCad E.D.A. 9.0.2Id: 1/1			
	Candidate direction: USB4105–class receptacle and low–capacitance USB ESD			Candidate direction: SK6805–EC15–class side emitters with current budget controls						
	Review scope: CC pull–downs, D+/D–, shield, return path, and connector mechanics			Review scope: Supply, 3.3 V data margin, direction, decoupling, and optical orientation						
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